

DRIE Project — Next Steps

1. Determine which existing wafers to retain

- Wafers **A1–A16** may be interesting to put back in the SEM for additional examination.
- Decide whether to retain any of these wafers before passing the remaining wafers to Fred for the ALD conformality project.

2. Determine the oxide-thinning method

- Discuss with Grace whether to use **RIE, ICP-7, vapor HF etcher, or BOE strip + re-deposit** to reduce the oxide toward the desired $\sim 4\text{ }\mu\text{m}$ thickness.
- Filmetrics measurements before thinning have already been completed. Use Filmetrics after thinning to verify thickness/uniformity.

Contact: Grace.

3. Prepare $\sim 4\text{ }\mu\text{m}$ oxide wafers

- Obtain suitable $4\text{ }\mu\text{m}$ oxide wafers, either directly or by thinning the existing oxide wafers using the method selected with Grace.
- Confirm the resulting oxide thickness before moving into lithography.

4. Decide on the lithography design

- Current masks include the **standard etch mask** and the **$1\text{ }\mu\text{m}$, $3\text{ }\mu\text{m}$, and $5\text{ }\mu\text{m}$ trench + $5\text{ }\mu\text{m}$ circular pillars + $5\text{ }\mu\text{m}$ square pillars** mask, which is in dry box C7.
- Decide whether to make a new design using the **MLA 150 direct-write** for better flexibility and better lithography resolution.
- Jonah and Labib are already working on similar MLA patterns and may provide a useful starting point for photo settings.

5. Run lithography development on the selected design

- Spin coat approximately **$7\text{ }\mu\text{m}$ resist** on the $4\text{ }\mu\text{m}$ oxide wafers.
- If using the MLA 150, establish photo settings that produce well-defined features.
- For $4\text{ }\mu\text{m}$ oxide, **$\sim 6\text{--}8\text{ }\mu\text{m}$ resist** should generally be sufficient, but this depends on oxide-etch performance.

Contact: Jonah / Labib for MLA/photo guidance.

6. Test Grace's ICP-7 oxide-etch recipe

- Before committing to DRIE, test the oxide etch properly on **ICP-7**. ICP-F is currently down.
- Test wafers can be re-used if the initial etch testing does not fully etch the oxide.
- The goal is to determine the time required to cleanly etch through the oxide without etching silicon significantly, and to check for **photoresist reticulation**.

Contact: Grace for the ICP-7 recipe/tool guidance.

7. Measure the oxide-etch results

- Use the **Keyence** to examine the oxide-etch results. **Nanospec** may also be useful for oxide or resist thickness measurements. Use SEM if necessary.
- Use the results to determine the appropriate oxide-etch time and whether the resist thickness or recipe needs adjustment.

8. Confirm DRIE tool viability before running the full process

- The proposed process increases deposition pressure and decreases etch pressure from the previous **40 / 30 / 30** condition to approximately **50 / 20 / 20**.
- Etch time is proposed at **2.5–3.5 seconds**, specified on the tool as a linear ramp, increased from the previous time of **2.5–3.0 seconds**.
- No change to platen power is currently recommended.
- Grace is unsure whether the tool is currently in good enough condition to attempt the proposed pressures. Process flow can be finalized while waiting for tool viability.

Contact: Grace.

9. Decision point: choose the DRIE run strategy

Option A — Full proposed recipe: use the proposed pressure changes (50 / 20 / 20) and increased etch time (2.5–3.5 s).
Option B — Preliminary / lower-risk test: retain the previous 40 / 30 / 30 pressures but test the increased etch time first.
Option C — Previous recipe control: run the MM_Deep_Std-3 recipe unchanged to provide a direct comparison against Mick's previous results.

- Also decide how many loops to use for the **3rd loop set**.
- Make this decision after the oxide-etch results and tool condition are understood.

10. Run DRIE and perform SEM metrology

- Once photo and oxide etch are qualified, proceed with the selected DRIE run strategy.
- Record the actual tool parameters and run conditions from **SPTS_DRIE_recipe.xlsx**; proposed changes for new runs are shown in blue.
- After etching, use **SEM** for metrology and compare results against the previous DRIE data.
- For wafer images, use the directional convention relative to North at the major flat: **C = Center**, **E = East**, **S = South**, etc.